Certified Least Values for OEIS A087354 at Indices 20–25

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Abstract

OEIS A087354 asks for the least perfect square that is a decimal concatenation of n nonzero perfect-square blocks, allowing repetitions. We determine the formerly unlisted value $a(20)$ and the values $a(22), \ldots, a(25)$ using exact-integer, exhaustive branch-and-bound certificates, and independently reproduce the reported value $a(21)$. The proof combines a digit-length reduction with a complete prefix-square interval test. This is an exact finite determination, *not* an all-index closed form or an analytic formula for general $a(n)$.

1 New exact values

Let $a(n)$ be the minimum positive square S whose usual decimal representation is a concatenation of exactly n positive squares. A one-digit nonzero square is in $D = \{1, 4, 9\}$, and a two-digit square is in $E = \{16, 25, 36, 49, 64, 81\}$.

Theorem 1. The following values are exact:

n	$a(n)$	$\sqrt{a(n)}$	digits
20	111125911149919941444	10541627538	21
21	444411911999914911441	21081079479	21
22	11199444494448114419449	105827427893	23
23	1449494491491491944481 44	380722272988	24
24	111142541419944919119 1449	1054241629893	25
25	41914991191911941194149136	6474178804444	26

The value at $n = 21$ was already noted on the OEIS page; the other five values are new determinations relative to the cited OEIS entry as accessed on October 22, 2026 [1].

2 Digit-length reduction and exact pruning

Every admissible S has at least n decimal digits because each square block has at least one digit. Each value displayed in Theorem 1 has at most $n + 1$ digits. Thus, to prove minimality, it suffices to examine lengths $L = n$ and $L = n + 1$ (and only $L = n$ for the 21-digit candidate at $n = 21$).

For $L = n$, every block is in D. For $L = n + 1$, precisely one block is in E and all remaining $n - 1$ blocks are in D: more than one extra digit or a block of three or more digits would exceed $n + 1$. Consequently the complete *candidate languages* are

$$\mathcal{W}_{n,n} = D^n, \qquad \mathcal{W}_{n,n+1} = \bigcup_{j=0}^{n-1} D^j E D^{n-1-j}.$$

Here each word means the decimal concatenation of its integer blocks. The two finite languages describe *all* possible candidates in the relevant length ranges, including repeated blocks.

Lemma 1 (Square-interval exclusion)**.** Let p be the integer represented by a prefix of d digits of a putative L-digit square, where $0 \le d \le L$. Every decimal completion of that prefix belongs to

$$I_{p,d,L} = [p10^{L-d}, (p + 1)10^{L-d} - 1] \cap \mathbb{Z}.$$

In particular, no completion is a square if

$$\left\lceil \sqrt{p10^{L-d}} \right\rceil > \left\lfloor \sqrt{(p+1)10^{L-d}-1} \right\rfloor .$$

For searching up to a known admissible upper bound U, the same assertion holds with the right endpoint replaced by $\min\{(p+1)10^{L-d}-1, U\}$.

Proof. The interval is exactly the set of nonnegative integers extending the prefix with $L-d$ decimal digits. An integer square exists in an integer interval $[A,B]$ precisely if the least integer square root at least $\sqrt{A}$ is no larger than the greatest integer square root at most $\sqrt{B}$. Intersecting with $[0,U]$ does not discard any candidate not exceeding U. □

3 Reproducible exhaustive certificate

For each n listed, the verifier starts with prefix $p=0$, $d=0$ digits, and $k=0$ blocks, and extends only by allowed blocks in D and (if one extra digit remains) E. Before visiting the children it applies Lemma 1; an empty square-root interval prunes the whole subtree. At $k=n$, a leaf is accepted only if $d=L$ and p is a perfect square. The integer upper bound U is set to the claimed value. A second branch-and-bound limit, the least square already found, can only remove larger values. Thus the search is exhaustive for all squares $\le U$ at each admissible digit length. The independent block-partition routine verifies the n-block representation of each proposed value, so each upper bound is valid. These facts prove Theorem 1 by finite exhaustion.

In particular, for $n=20$ the first witness is

$$\begin{aligned} 10541627538^2 = 1 \parallel 1 \parallel 1 \parallel 1 \parallel 25 \parallel 9 \parallel 1 \parallel 1 \parallel 1 \parallel 4 \\ \parallel 9 \parallel 9 \parallel 1 \parallel 9 \parallel 9 \parallel 4 \parallel 1 \parallel 4 \parallel 4 \parallel 4. \end{aligned}$$

There are no eligible 20-digit squares, and every 21-digit eligible square smaller than this value is excluded. A complete per-length search trace and the remaining factorizations are generated by `verify_minima.py` and `verify_minima.cpp`.

The C++ implementation is an independent fast audit with native 128-bit integers; a floating-point square root is used *only* as a provisional estimate and is adjusted by exact integer-square comparisons before any accept/reject decision. The Python version uses `math.isqrt` throughout, without floating point. Both implementations agree on all six values and all complete search counts. Running the Python script with `-extended` includes the more expensive $n=25$ case.

Scope. This note does not prove any all-index formula for $a(n)$. For each index treated, the bound $L \le n+1$ follows from an explicit witness; we do not assume it for arbitrary n. It remains open here whether this bound, or a tractable exact closed formula, holds at every index. The earlier separate existence manuscript proves that all indices are defined but not their minima.

References

[1] OEIS Foundation Inc., *A087354: Smallest square which is the concatenation of n nonzero squares*, https://oeis.org/A087354, accessed October 22, 2026.